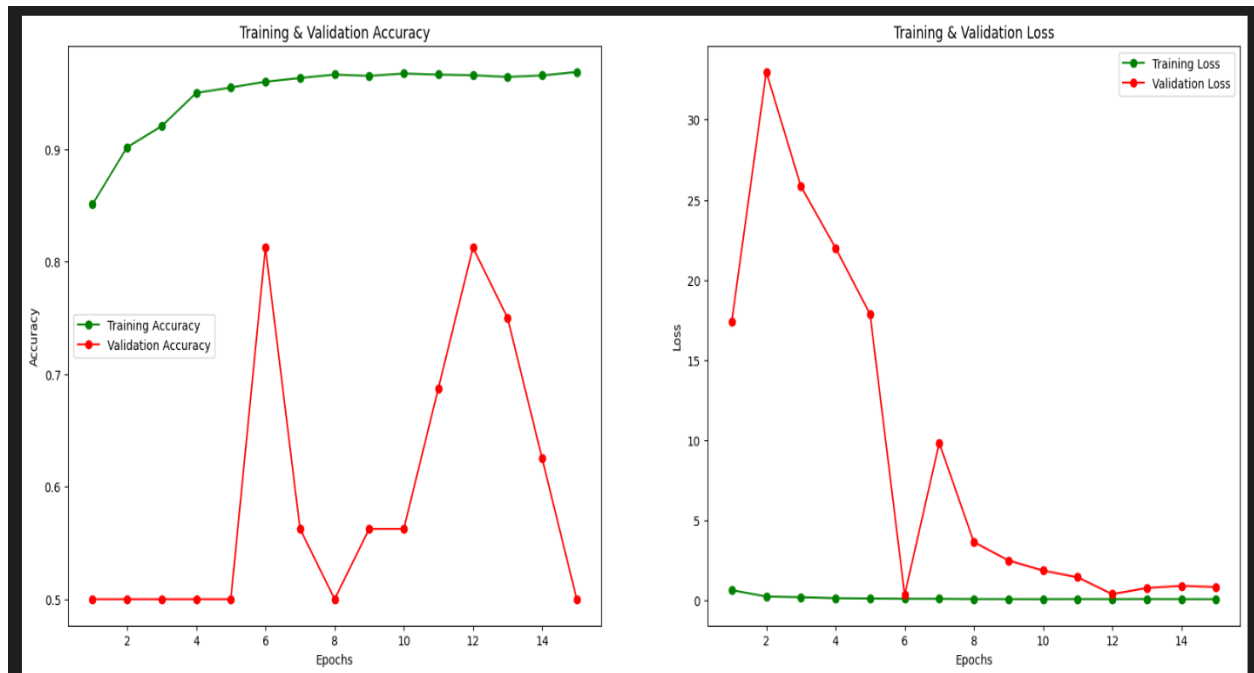


Pneumonia Detection – Model Outputs

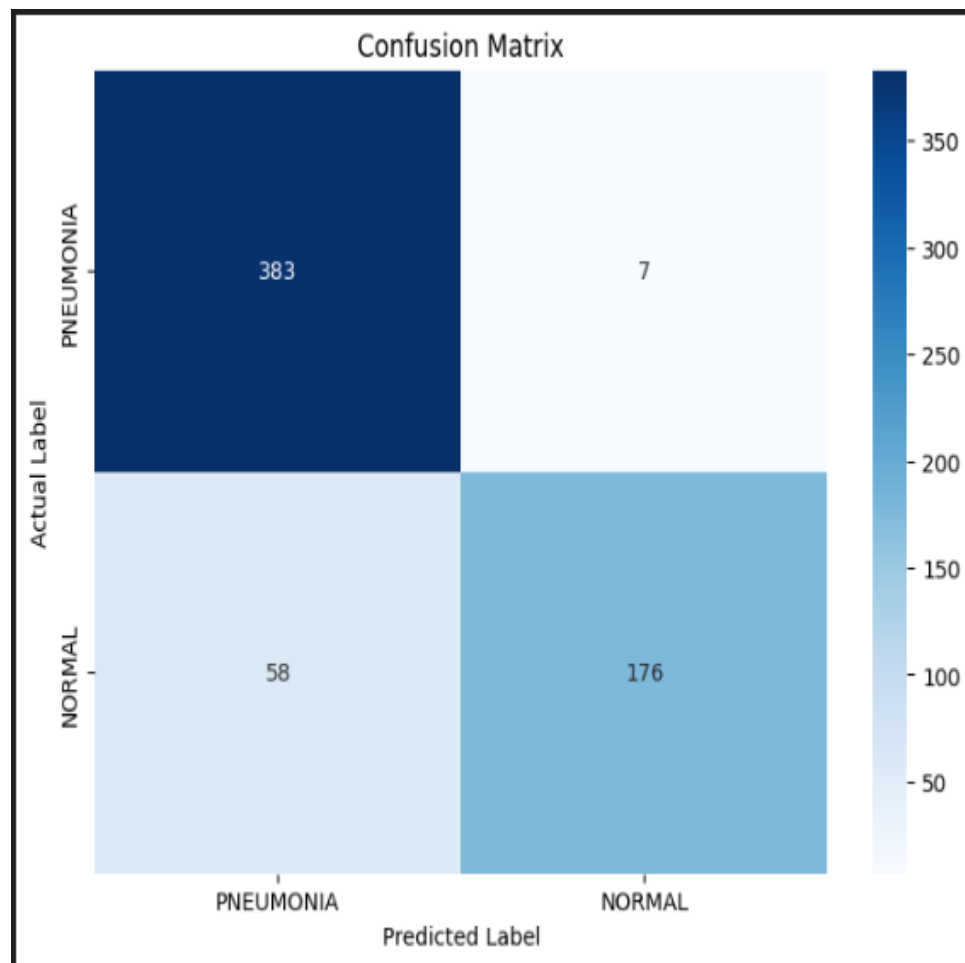
1. Training & Validation Performance

Training and validation accuracy and loss curves across 15 epochs.



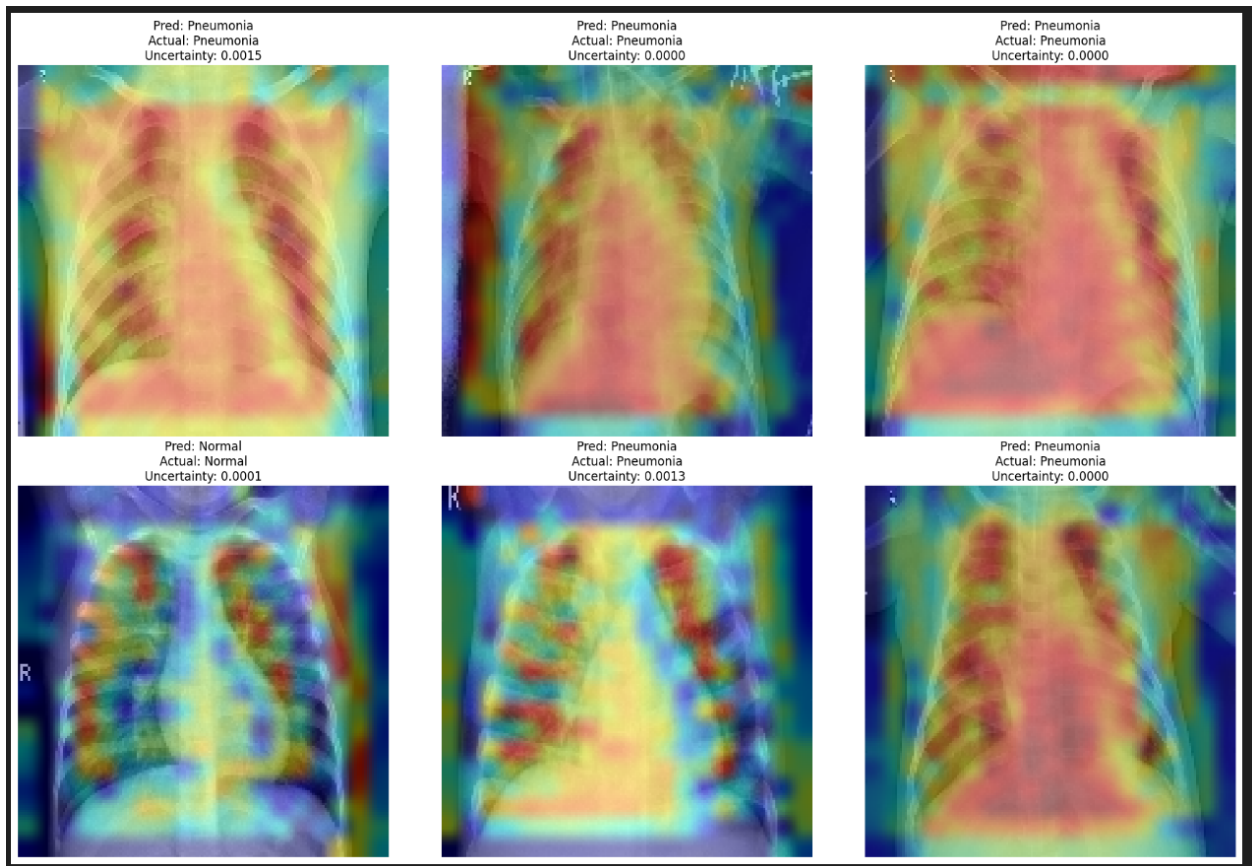
2. Test Set Evaluation

Confusion matrix showing model performance on Normal and Pneumonia classes.



3. Explainability – Grad-CAM

Grad-CAM visualizations highlighting lung regions influencing predictions.



4. Classification Report (Monte Carlo Dropout)

Precision, recall, F1-score, and accuracy under uncertainty-aware inference.

```
20/20 ————— 1s 8ms/step - accuracy: 0.8665 - loss: 0.2820  
Loss of the model is - 0.26360949873924255  
20/20 ————— 0s 8ms/step - accuracy: 0.8665 - loss: 0.2820  
Accuracy of the model is - 88.14102411270142 %
```

Classification Report (with Monte Carlo Dropout):

	precision	recall	f1-score	support
Pneumonia (Class 0)	0.87	0.98	0.92	390
Normal (Class 1)	0.96	0.75	0.84	234
accuracy			0.90	624
macro avg	0.92	0.87	0.88	624
weighted avg	0.90	0.90	0.89	624